FastSpeech 2 English and Telugu

Technical report | Separate assignment-trained acoustic models

Evidence cut-off: 2026-11-06 12:05 UTC

Two independent models use approximately 5 h training and 1 h held-out validation each. Bounded training stopped at English step 8,842 and Telugu step 7,451. Final inference uses English 8842 with sentence context and ordinary energy conditioning (SIL zeroing OFF), and unchanged Telugu 2000. E1/E2a/E2b use contextual English input; E3/E4 use the trained-word fallback. All ten real WAVs and requested plots exist. This report is objective-only: no native listening, satisfactory pronunciation, naturalness or clean blind-test result is claimed.

Evidence boundary

This report is generated from production split manifests, saved training and validation logs, the selected checkpoints used for inference, and objective measurements of the ten final WAV files. The builder verifies current file hashes before rendering. It does not substitute development smoke runs or benchmark checkpoints for production evidence.

Model	Train / validation	Latest / selected step	Final samples
English	5.0001 h / 1.0010 h	8,842 / 8,842	E1, E2a, E2b, E3, E4
Telugu	5.0015 h / 1.0039 h	7,451 / 2,000	T1, T2a, T2b, T3, T4

Assessment coverage: 0 WAVs with explicitly recorded listening; 10 objective-only reviews. Objective-only entries leave perceived speech quality and pronunciation unassessed. No MOS, intelligibility score, or native-speaker verdict is inferred.

Deliverables checked

Two separate acoustic checkpoints; ten generated WAVs; thirty duration/F0/energy plots; four loss plots; ten raw prediction archives; two current analysis.json files; exact source sentence JSON files; and this report. All reported timing and signal values are measurements of those artifacts.

Reading guide

Architecture and adaptations; data, splits and hyperparameters; production loss curves; individual sample reviews; cross-language comparison and limitations; reproduction; source and license records.

Architecture and reference adaptations

Reference: ming024/FastSpeech2 at commit d4e79eb52e8b01d24703b2dfc0385544092958f3 [1]. The vendored encoder, variance adaptor, length regulator, decoder, postnet and reference loss are retained. The acoustic models start without pretrained TTS acoustic weights. The separate pretrained universal HiFi-GAN decodes predicted mel spectrograms to waveforms [2,3].

Stage	Implemented behavior
Text front end	NFC and whitespace normalization; eSpeak NG IPA tokens; English AM/PM time-suffix expansion; original assignment text retained.
Encoder	Learned IPA embedding and positional encoding, followed by FFT blocks. Independent language vocabularies and weights.
Variance adaptor	Phone-level log-duration, pitch and energy predictors; pitch/energy embeddings; duration-based length regulation.
Decoder / postnet	FFT decoder predicts 80-bin mel frames; postnet adds a mel residual. Losses: mel, postnet mel, duration, pitch, energy.
Waveform synthesis	Bundled pretrained universal HiFi-GAN at 22,050 Hz. Its weights are not an assignment-trained acoustic model.

Project adaptations

tts/modeling.py replaces the reference symbol embedding with each prepared IPA vocabulary and sets device-aware helper allocations. Shared modules expose separate English and Telugu entry points, configurations, statistics, speaker maps, manifests, checkpoints and outputs. The training loop adds split-integrity gates, checkpoint contract hashes, reproducible batch ordering, interruption resume, gradient clipping and separate validation. Validation averages per-utterance reference losses; training uses the reference masked batch reduction.

Language-specific alignment and conditioning

English uses CUPE-3-en alignment. Svarah's available export lacks verified speaker IDs, so hashed demographic profiles condition the multi-speaker model. These are proxy groups, not verified people. Telugu uses one female voice and a single-speaker acoustic model. CUPE-3-mls7 has no Telugu training; its alignment is zero-shot. Telugu-only acoustic mappings approximate /palatalization/ and two aspirated consonants while preserving original IPA labels as model inputs. The acoustic mapping collapses some Telugu distinctions and does not validate their pronunciation [6-8].

Front-end and inference controls

Training used the original word-level English front end. The sentence-context inference option can change the 'conduct' vowel by context; lexical stress is not encoded, so it is not a complete stress or homograph-quality solution. Requested/used modes and fallback status are recorded in each inference manifest. The visible 'rhyth m', 'andaroo', and Telugu internal-space challenge remain in the exact source text.

Recorded English modes: {'sentence_context_v1': 3, 'trained_word_level': 2}. Zero-SIL-energy conditioning was applied to 0/5 English and 0/5 Telugu samples. When enabled, only the energy values fed to the decoder are changed at SIL positions; raw energy predictions and checkpoint training/validation metrics remain unchanged.

Data preparation, splits and hyperparameters

Svarah [4] is repurposed from its original ASR test partition into new TTS train/validation subsets; this is not the standard benchmark evaluation. The imported files measure 9.6144 h at 16 kHz. IndicTTS Telugu [5] uses the female subset of the first two training shards; the 1-30 s candidate pool measures 13.3541 h at 16 kHz. The Telugu card's 48 kHz/8.74 h summary contradicts the downloaded files. Candidate pools are distinct from the accepted splits below.

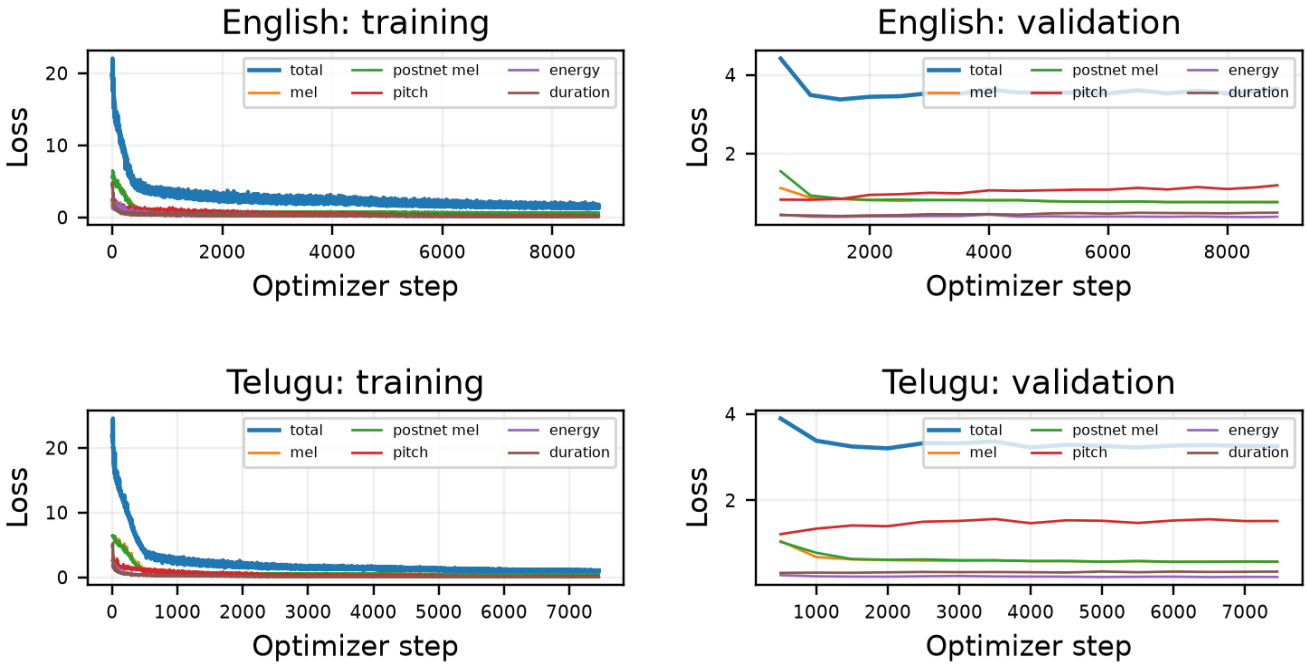
Measured preparation	English	Telugu
Train utterances / hours	3047 / 5.00011	1177 / 5.00148
Validation utterances / hours	615 / 1.00104	241 / 1.00392
Accepted / rejected candidates	3971 / 383	1541 / 46
Candidate prefix inspected	4354 rows	1587 rows
Train / validation conditioning IDs	114 / 106	1 / 1
Vocabulary incl. padding / seed	60 / 20261105	49 / 20261105
QA max phone s / min confidence	1.0 / 0.2	1.0 / 0.2

Eligible 1-30 s source rows are sorted by ID and shuffled with seed 20261105. English takes whole recordings through the first 7.5 h; Telugu retains the shuffled female pool. The accepted scan stops at 6.5 h or candidate exhaustion. This is followed by seeded allocation of whole connected groups sharing normalized text or audio hashes. Utterances are not arbitrarily cropped. Train/validation IDs, audio, normalized text and feature identities are disjoint. Validation conditioning IDs must occur in training. Pitch and energy statistics are fitted only on training targets. Vocabulary coverage includes the prescribed test text, but no test recording or acoustic target.

Actual checkpoint setting	English	Telugu
FFT encoder / decoder / hidden / heads	4 / 6 / 256 / 2	4 / 6 / 256 / 2
FFT filter / kernels / dropout	1024 / [9, 1] / 0.2	1024 / [9, 1] / 0.2
Variance filter / kernel / dropout	256 / 3 / 0.5	256 / 3 / 0.5
Pitch / energy bins and quantization	256; linear / linear	256; linear / linear
Batch / precision / length bucket	16 / bf16 / 64	16 / bf16 / 64
Peak learning rate / warmup steps	0.001 / 4000	0.001 / 4000
Gradient clip / weight decay	1.0 / 0	1.0 / 0
Validation interval / saved config target	500 / 50000	500 / 50000
Multispeaker / maximum frames	True / 3000	False / 3000

AdamW uses betas (0.9, 0.98), epsilon 1e-9. If warmup is enabled, learning rate rises linearly then falls with inverse square-root step. Features: resample to 22,050 Hz; 1,024-sample FFT/window; 256-sample hop; 80 natural-log magnitude Slaney mel bins, 0-8,000 Hz. WORLD DIO/StoneMask F0 is interpolated through unvoiced regions before phone averaging. Energy is the linear STFT magnitude norm, then phone averaged. Acoustic QA rejects backend flags, speech-phone spans above the recorded threshold and low mean confidence; these heuristic gates are not human alignment verification.

Training loss and held-out validation



Curves are redrawn from the complete production logs in checkpoint/logs; full-size original plots remain in Results/<language>/{training_loss,validation_loss}.png. Loss values are dimensionless criterion values, not a speech-quality scale. Training and validation aggregation differ.

Model	First / last train total	First / last validation total	Minimum validation / step
English	19.732 / 1.557	4.413 / 3.643	3.374 / 1,500
Telugu	21.928 / 1.059	3.894 / 3.252	3.204 / 2,000

English: Training total falls from 19.7317 to 1.5567; the last-100-step mean is 1.4998. Held-out total falls from 4.4130 to 3.6431, with minimum 3.3738 at step 1500. The bounded run ends at step 8,842; this is not a claim that the configured target or satisfactory speech quality was reached. Instability/overfitting: Training keeps improving after the validation minimum; final validation total is 0.2693 above that minimum. This plateau/gap is compatible with limited generalization or overfitting, not proof of a single cause. Training uses masked batch reduction while validation averages per utterance. Final validation pitch loss is 1.1972; the pitch criterion includes SIL and zero-duration positions, so it is not a voiced-speech F0 error score.

Telugu: Training total falls from 21.9279 to 1.0588; the last-100-step mean is 0.9674. Held-out total falls from 3.8938 to 3.2523, with minimum 3.2036 at step 2000. The bounded run ends at step 7,451; this is not a claim that the configured target or satisfactory speech quality was reached. Instability/overfitting: Training keeps improving after the validation minimum; final validation total is 0.0487 above that minimum. This plateau/gap is compatible with limited generalization or overfitting, not proof of a single cause. Training uses masked batch reduction while validation averages per utterance. Final validation pitch loss is 1.5208; the pitch criterion includes SIL and zero-duration positions, so it is not a voiced-speech F0 error score.

An endpoint reduction or validation minimum alone does not establish convergence. Quality, undertraining and overfitting conclusions remain bounded by the recorded trajectory and sample assessment coverage. best.pt is always retained as the minimum-total-validation-loss baseline. An explicit selection.json may designate selected.pt for inference; otherwise inference uses best.pt.

English: inference selected.pt step 8,842; minimum-total best.pt step 1,500; latest.pt / final logged step 8,842; last training invocation target 10,000, completed optimizer updates during that invocation 842. A saved configuration target is not a claim that all of its planned steps were run.

Telugu: inference selected.pt step 2,000; minimum-total best.pt step 2,000; latest.pt / final logged step 7,451; last training invocation target 10,000, completed optimizer updates during that invocation 451. A saved configuration target is not a claim that all of its planned steps were run.

Checkpoint selection and validation probes

Earlier ASR diagnostics on the assignment prompts motivated this review. The subsequent original frozen comparison used three held-out validation prompts and two supplied checkpoints per language. Further English inference/checkpoint experiments are explicitly post-hoc; their outcome is not covered by the original two-candidate protocol. None is a pristine blind final test, a final-test quality score, or a human listening assessment.

English

Checkpoint: selected.pt, step 8,842. Select final checkpoint 8842 for improved fixed-validation recognition (WER 15.38%, CER 11.19% versus step-7000 38.46%/30.60%) and enable sentence context to correct the known conduct verb-vowel target. Retain normal energy and duration inference. Although SIL-energy-zero improved the three validation probes, it worsened assignment recognition and was rejected. Neither a perfect speech-quality claim nor a passed old timing gate is made.

Protocol: Exploratory post-hoc refinement, not a blind test: initial assignment-ASR concerns motivated fixed-validation diagnostics and causal ablations. Both assignment and validation outputs were examined. Earlier selection records are preserved as selection_v1.json and selection_v2.json. Duration scaling and zero-SIL-energy conditioning were tested and rejected; final controls remain 1.0 and raw energy is used. Context corrects vowel targets but stress markers remain unencoded.

Frozen comparison outcome: inconclusive; no validated-quality winner was established.

Post-hoc preference: both the checkpoint and inference recipe may differ from the earlier experiment. Any measured improvement cannot be attributed to additional training alone. The failed duration gate remains a failure, and raw full-validation losses are not recomputed or replaced by these synthesis diagnostics.

Reported comparison	WER	CER	Generated/source duration	Screening checks
minimum total step 1,500	0.692	0.455	0.402-0.858	Audio integrity: True; duration gate: FAIL
later fixed step 8,842	0.154	0.112	0.490-0.919	Audio integrity: True; duration gate: FAIL
Source-recording ASR control	0.077	0.015	Reference audio	Control

WER and CER above are proportions, micro-averaged across only these three validation prompts; duration ratios report their minimum and maximum. The predeclared duration gate requires every ratio in [0.5, 2.0]. Audio integrity is a separate flag and does not override a failed duration gate or certify speech quality. Source-recording controls help reveal recognizer errors and text-normalization effects. None of these values establish pronunciation correctness or perceived speech quality.

en_test-00000-of-00003_000970: Brazil, Cambodia, Japan, Spain, Greece.

en_test-00000-of-00003_002008: eat seasonal fruits and use these flowers for worshipping and so on.

en_test-00001-of-00003_001541: Especially this Yaksha Gana and its various versions are

Decision: posthoc validation improvement. Full record: English/checkpoints/selection.json. Frozen prompt SHA256: ece858ca512c3727fc795406b38f4b839fd8e2e66695966fee9c85c4f6b282ed

Preserved original decision: English/checkpoints/selection_v1.json selected step 7000 with decision later best available failed quality gate. Its original metrics and failed/inconclusive gate outcome remain historical evidence; the new synthesis recipe does not rewrite them.

The later experimental recipe is also preserved in English/checkpoints/selection_v2.json (step 8842). Its metrics and decision are historical; they do not replace the final recipe or turn rejected conditioning changes into improvements.

Checkpoint selection | Telugu

Checkpoint: selected.pt, step 2,000. The fixed validation comparison is inconclusive: later WER worsens while CER improves. Retain the original minimum-total-validation checkpoint. This choice does not establish native-listener speech quality.

Protocol: Earlier assignment-ASR diagnostics motivated this post-hoc review; the assignment set is not a pristine untouched test. Exactly three validation prompts and two checkpoints were then fixed before later-candidate evaluation. The frozen rule remains inconclusive. No native listening was performed. English artifact preference is an explicitly disclosed best-available decision despite the failed duration gate; no threshold or score was changed.

Frozen comparison outcome: inconclusive; no validated-quality winner was established.

Reported comparison	WER	CER	Generated/source duration	Screening checks
minimum total step 2,000	0.735	0.121	0.912-0.967	Audio integrity: True; duration gate: pass
later fixed step 6,000	0.794	0.102	0.914-0.981	Audio integrity: True; duration gate: pass
Source-recording ASR control	0.676	0.087	Reference audio	Control

WER and CER above are proportions, micro-averaged across only these three validation prompts; duration ratios report their minimum and maximum. The predeclared duration gate requires every ratio in [0.5, 2.0]. Audio integrity is a separate flag and does not override a failed duration gate or certify speech quality. Source-recording controls help reveal recognizer errors and text-normalization effects. None of these values establish pronunciation correctness or perceived speech quality.

te_train-00000-of-00004_000018: exact Telugu source in provenance/Telugu_validation_probes.json.

te_train-00000-of-00004_000044: exact Telugu source in provenance/Telugu_validation_probes.json.

te_train-00000-of-00004_000156: exact Telugu source in provenance/Telugu_validation_probes.json.

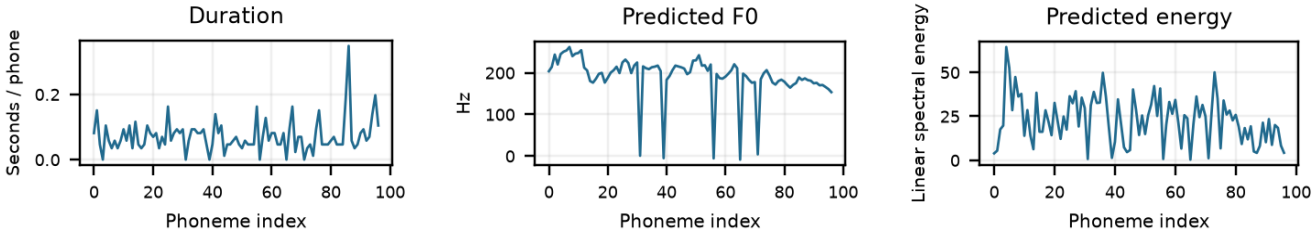
Decision: baseline retained inconclusive. Full record: Telugu/checkpoints/selection.json. Frozen prompt SHA256: 3a6d89d4f99c3b51b969a7b6f2b20c4d56d7ac699d9f9c3e9245fdaa6181047f

Individual samples | E1 and E2a

E1 | English | objective only

Source text: "The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."

WAV 6.978 s; RMS -24.85 dBFS; peak 0.428; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/86; longest non-SIL duration 0.197 s at phone index 95; longest SIL run 0.348 s.



Non-SIL predicted F0 mean 202.4 Hz; range 160.8-261.9 Hz. Non-SIL predicted energy mean 24.978. Waveform F0 status: measured.

Duration: Total 6.978 s; SIL 0.557 s (8.0%). Longest SIL run 0.348 s at phone index 86. All 6 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 160.8-261.9 Hz, mean 202.4. Its 24.0 Hz standard deviation is the broadest English contour here; that does not establish appropriate rare-word stress. WORLD voiced-WAV mean 200.1 Hz is a different estimate, not a target error.

Energy: Raw non-SIL energy 4.16-64.40, mean 24.98. Raw and decoder-conditioning energy are identical; SIL zeroing is OFF. SIL minimum 0.33; these spectral units are not loudness.

Limitations: Strict ASR WER 40.0%, CER 10.2%; the hypothesis begins 'then disestablish humanitarianism'. This still mismatches the rare source word. Sentence-context input and better character agreement do not verify phonetic accuracy or naturalness.

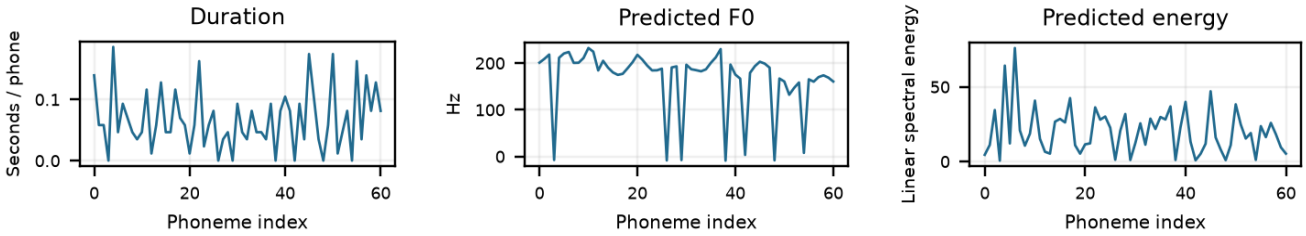
Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/English/test_generated_audios/E1.wav; predictions/E1.npz; variance_adaptor_plots/{duration,pitch,energy}/E1.png.

E2a | English | objective only

Source text: "The orchestra's conduct during the ceremony was praised by the audience."

WAV 4.087 s; RMS -23.53 dBFS; peak 0.489; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/49; longest non-SIL duration 0.186 s at phone index 4; longest SIL run 0.139 s.



Non-SIL predicted F0 mean 191.0 Hz; range 146.7-232.2 Hz. Non-SIL predicted energy mean 23.759. Waveform F0 status: measured.

Duration: Total 4.087 s; SIL 0.255 s (6.2%). Longest SIL run 0.139 s at phone index 0. All 7 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 146.7-232.2 Hz, mean 191.0. Its 191.0 Hz mean is close to E2b; contextual vowel targets differ, but no stress-pattern conclusion follows. WORLD voiced-WAV mean 189.1 Hz is a different estimate, not a target error.

Energy: Raw non-SIL energy 5.07-75.97, mean 23.76. Raw and decoder-conditioning energy are identical; SIL zeroing is OFF. SIL minimum 0.50; these spectral units are not loudness.

Limitations: Strict ASR WER 27.3%, CER 18.3%; the hypothesis substitutes 'artist', 'contact' and 'placed'. Contextual input changes the noun-vowel target, but this does not establish a correct audible noun form or encoded stress.

Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/English/test_generated_audios/E2a.wav; predictions/E2a.npz; variance_adaptor_plots/{duration,pitch,energy}/E2a.png.

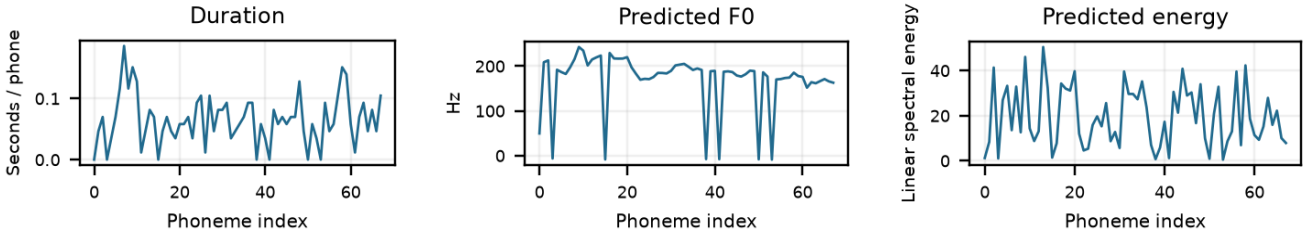
Plots use the actual phone-level predictions, indexed from zero; each includes SIL entries. Full phoneme labels are available in the corresponding PNG and .npz. Predicted F0 and energy are denormalized with training statistics. The energy panel retains the raw prediction and adds a dashed decoder-conditioning overlay only when the saved SIL control was active. These are not waveform F0 or RMS.

Individual samples | E2b and E3

E2b | English | objective only

Source text: "The researchers will conduct an experiment to evaluate the proposed method."

WAV 4.423 s; RMS -25.13 dBFS; peak 0.484; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/56; longest non-SIL duration 0.186 s at phone index 7; longest SIL run 0.104 s.



Non-SIL predicted F0 mean 192.0 Hz; range 161.6-242.6 Hz. Non-SIL predicted energy mean 22.967. Waveform F0 status: measured.

Duration: Total 4.423 s; SIL 0.174 s (3.9%). Longest SIL run 0.104 s at phone index 67. All 7 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 161.6-242.6 Hz, mean 192.0. Its 192.0 Hz mean is close to E2a; the verified input-vowel distinction is separate from pitch and audible stress. WORLD voiced-WAV mean 192.3 Hz is a different estimate, not a target error.

Energy: Raw non-SIL energy 4.54-50.41, mean 22.97. Raw and decoder-conditioning energy are identical; SIL zeroing is OFF. SIL minimum 0.46; these spectral units are not loudness.

Limitations: ASR WER/CER are 0/0 on normalized text. The contextual phoneme input now distinguishes the verb vowel from E2a's noun vowel; lexical stress is not encoded. Exact text recognition still does not establish pronunciation or naturalness.

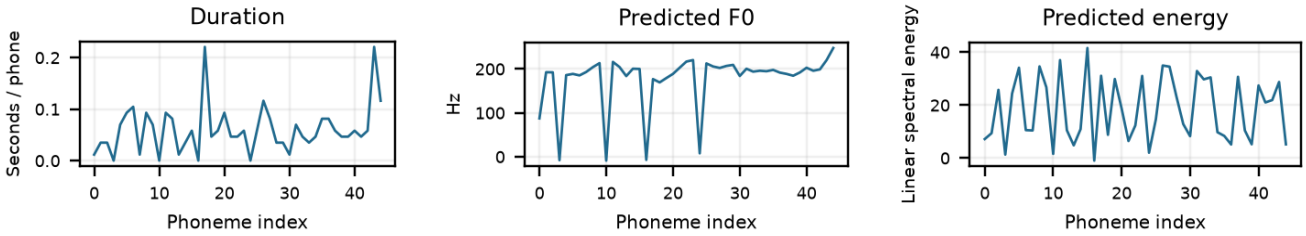
Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/English/test_generated_audios/E2b.wav; predictions/E2b.npz; variance_adaptor_plots/{duration,pitch,energy}/E2b.png.

E3 | English | objective only

Source text: "The rhyth m of the orchestra changed unexpectedly."

WAV 2.717 s; RMS -25.18 dBFS; peak 0.302; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/36; longest non-SIL duration 0.221 s at phone index 17; longest SIL run 0.116 s.



Non-SIL predicted F0 mean 198.0 Hz; range 169.2-220.1 Hz. Non-SIL predicted energy mean 21.810. Waveform F0 status: measured.

Duration: Total 2.717 s; SIL 0.163 s (6.0%). Longest SIL run 0.116 s at phone index 44. All 4 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 169.2-220.1 Hz, mean 198.0. Its 11.9 Hz standard deviation is the narrowest English contour here; no naturalness judgment follows. WORLD voiced-WAV mean 197.8 Hz is a different estimate, not a target error.

Energy: Raw non-SIL energy 5.08-41.57, mean 21.81. Raw and decoder-conditioning energy are identical; SIL zeroing is OFF. The SIL minimum -1.02 is a nonphysical unconstrained prediction.

Limitations: Strict ASR WER 50.0%, CER 23.8%; the hypothesis contains 'rhythm' and 'estrogen'. The exact reference retains 'rhyth m'. Contextual word-count mismatch triggered the trained word-level fallback; this is not a verified phonetic diagnosis.

Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/English/test_generated_audios/E3.wav; predictions/E3.npz; variance_adaptor_plots/{duration,pitch,energy}/E3.png.

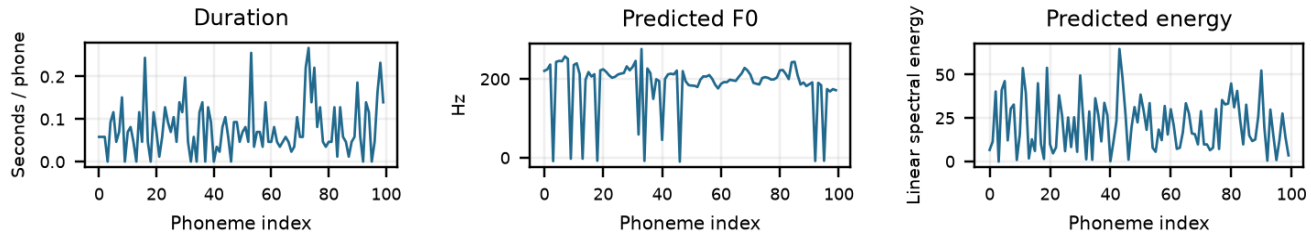
Plots use the actual phone-level predictions, indexed from zero; each includes SIL entries. Full phoneme labels are available in the corresponding PNG and .npz. Predicted F0 and energy are denormalized with training statistics. The energy panel retains the raw prediction and adds a dashed decoder-conditioning overlay only when the saved SIL control was active. These are not waveform F0 or RMS.

Individual samples | E4 and T1

E4 | English | objective only

Source text: "The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."

WAV 7.848 s; RMS -23.20 dBFS; peak 0.543; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/81; longest non-SIL duration 0.255 s at phone index 53; longest SIL run 0.267 s.



Non-SIL predicted F0 mean 209.3 Hz; range 168.1-275.9 Hz. Non-SIL predicted energy mean 25.076. Waveform F0 status: measured.

Duration: Total 7.848 s; SIL 0.615 s (7.8%). Longest SIL run 0.267 s at phone index 73. All 11 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 168.1-275.9 Hz, mean 209.3. Its 209.3 Hz mean is the highest English value here; intended emphasis and date pronunciation are unverified. WORLD voiced-WAV mean 205.9 Hz is a different estimate, not a target error.

Energy: Raw non-SIL energy 5.41-64.68, mean 25.08. Raw and decoder-conditioning energy are identical; SIL zeroing is OFF. The SIL minimum -0.23 is a nonphysical unconstrained prediction.

Limitations: Strict ASR WER 27.8%, CER 10.0%; the hypothesis includes 'until November' and 'Pandoro'. Dates/time and 'andaroo' remain literal references. Contextual word-count mismatch triggered word-level fallback; number/code-switch correctness is unassessed.

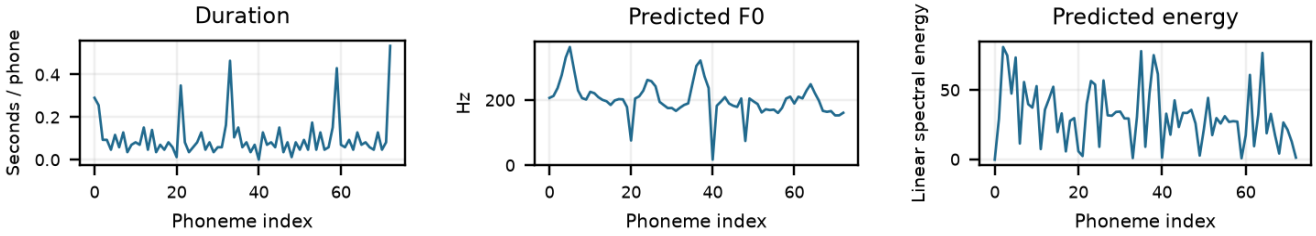
Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/English/test_generated_audios/E4.wav; predictions/E4.npz; variance_adaptor_plots/{duration,pitch,energy}/E4.png.

T1 | Telugu | objective only

Exact Telugu source text: Telugu/test_samples.json, entry T1. The unchanged UTF-8 source is authoritative; complex-script shaping was not validated in this Roman-script PDF.

WAV 7.465 s; RMS -23.22 dBFS; peak 0.401; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/66; longest non-SIL duration 0.348 s at phone index 21; longest SIL run 0.534 s.



Non-SIL predicted F0 mean 209.9 Hz; range 152.5-362.6 Hz. Non-SIL predicted energy mean 34.327. Waveform F0 status: measured.

Duration: Total 7.465 s; SIL 1.741 s (23.3%). Longest SIL run 0.534 s at phone index 72. All 1 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 152.5-362.6 Hz, mean 209.9. Its 362.6 Hz maximum and 43.3 Hz standard deviation are the largest Telugu values in this set. WORLD voiced-WAV mean 184.3 Hz is a different estimate, not a target error.

Energy: Non-SIL spectral energy 2.21-80.89, mean 34.33. SIL minimum -0.37 is nonphysical after denormalization; the regression is unconstrained. These units are not loudness or waveform RMS.

Limitations: ASR WER 166.7%, CER 21.0%: 10 word edits against 6 reference words, with 11 hypothesis words. Insertions/segmentation can make WER exceed 100%. This is not a native-speaker or phonetic assessment.

Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/Telugu/test_generated_audios/T1.wav; predictions/T1.npz; variance_adaptor_plots/{duration,pitch,energy}/T1.png.

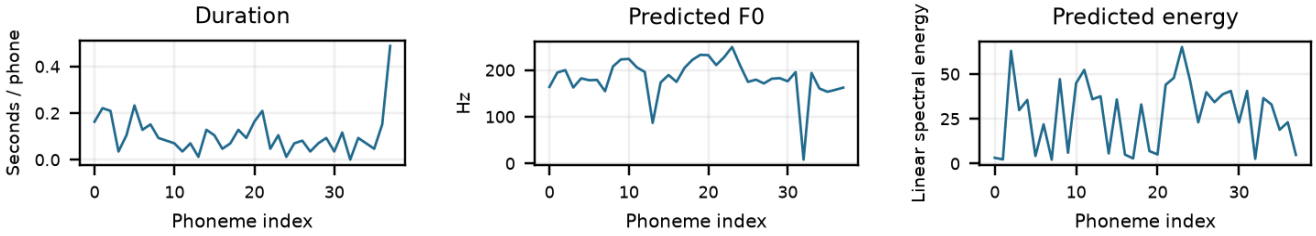
Plots use the actual phone-level predictions, indexed from zero; each includes SIL entries. Full phoneme labels are available in the corresponding PNG and .npz. Predicted F0 and energy are denormalized with training statistics. The energy panel retains the raw prediction and adds a dashed decoder-conditioning overlay only when the saved SIL control was active. These are not waveform F0 or RMS.

Individual samples | T2a and T2b

T2a | Telugu | objective only

Exact Telugu source text: Telugu/test_samples.json, entry T2a. The unchanged UTF-8 source is authoritative; complex-script shaping was not validated in this Roman-script PDF.

WAV 4.052 s; RMS -24.30 dBFS; peak 0.479; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/31; longest non-SIL duration 0.232 s at phone index 5; longest SIL run 0.488 s.



Non-SIL predicted F0 mean 192.1 Hz; range 153.3-249.2 Hz. Non-SIL predicted energy mean 30.214. Waveform F0 status: measured.

Duration: Total 4.052 s; SIL 0.917 s (22.6%). Longest SIL run 0.488 s at phone index 37. All 1 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 153.3-249.2 Hz, mean 192.1. Its 192.1 Hz mean is the lowest Telugu value here; the comparison is not a controlled semantic experiment. WORLD voiced-WAV mean 165.6 Hz is a different estimate, not a target error.

Energy: Non-SIL spectral energy 2.25-65.03, mean 30.21. SIL minimum is 2.10. These units are not loudness or waveform RMS.

Limitations: Strict ASR WER 66.7%, CER 10.7% (4 word edits; 3 character edits). The smaller character discrepancy illustrates sensitivity to Telugu word boundaries as well as recognition differences; no correctness claim follows.

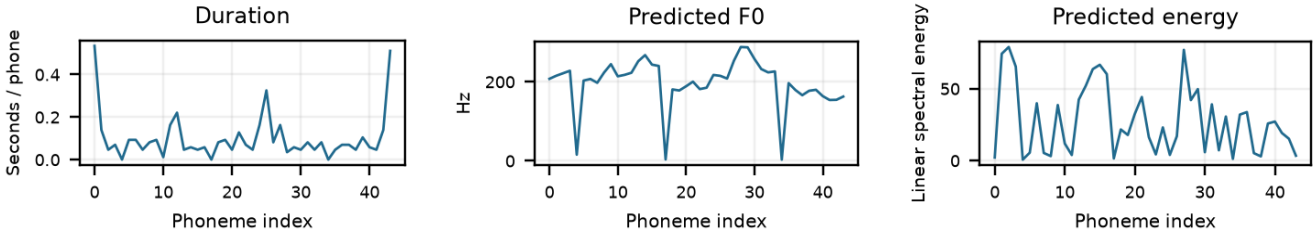
Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/Telugu/test_generated_audios/T2a.wav; predictions/T2a.npz; variance_adaptor_plots/{duration,pitch,energy}/T2a.png.

T2b | Telugu | objective only

Exact Telugu source text: Telugu/test_samples.json, entry T2b. The unchanged UTF-8 source is authoritative; complex-script shaping was not validated in this Roman-script PDF.

WAV 4.435 s; RMS -24.01 dBFS; peak 0.447; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/37; longest non-SIL duration 0.221 s at phone index 12; longest SIL run 0.534 s.



Non-SIL predicted F0 mean 212.4 Hz; range 153.8-287.8 Hz. Non-SIL predicted energy mean 32.112. Waveform F0 status: measured.

Duration: Total 4.435 s; SIL 1.382 s (31.2%). Longest SIL run 0.534 s at phone index 0. All 3 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 153.8-287.8 Hz, mean 212.4. Its 212.4 Hz mean exceeds T2a's 192.1 Hz; different context prevents attributing this to meaning alone. WORLD voiced-WAV mean 172.3 Hz is a different estimate, not a target error.

Energy: Non-SIL spectral energy 2.91-78.92, mean 32.11. SIL minimum is 0.64. These units are not loudness or waveform RMS.

Limitations: Strict ASR WER 33.3%, CER 0.0%: two word edits despite identical normalized non-space characters. The hypothesis splits a reference word. This spacing effect and zero CER do not establish correct speech or stress.

Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/Telugu/test_generated_audios/T2b.wav; predictions/T2b.npz; variance_adaptor_plots/{duration,pitch,energy}/T2b.png.

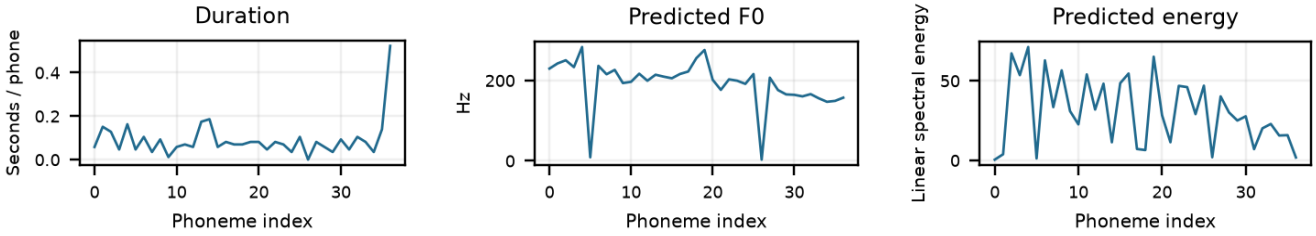
Plots use the actual phone-level predictions, indexed from zero; each includes SIL entries. Full phoneme labels are available in the corresponding PNG and .npz. Predicted F0 and energy are denormalized with training statistics. The energy panel retains the raw prediction and adds a dashed decoder-conditioning overlay only when the saved SIL control was active. These are not waveform F0 or RMS.

Individual samples | T3 and T4

T3 | Telugu | objective only

Exact Telugu source text: Telugu/test_samples.json, entry T3. The unchanged UTF-8 source is authoritative; complex-script shaping was not validated in this Roman-script PDF.

WAV 3.355 s; RMS -23.13 dBFS; peak 0.448; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/31; longest non-SIL duration 0.174 s at phone index 13; longest SIL run 0.522 s.



Non-SIL predicted F0 mean 205.3 Hz; range 146.8-283.2 Hz. Non-SIL predicted energy mean 35.466. Waveform F0 status: measured.

Duration: Total 3.355 s; SIL 0.824 s (24.6%). Longest SIL run 0.522 s at phone index 36. All 1 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 146.8-283.2 Hz, mean 205.3. The observed range describes the model contour only; no intended lexical or phrase stress is verified. WORLD voiced-WAV mean 189.0 Hz is a different estimate, not a target error.

Energy: Non-SIL spectral energy 3.81-71.20, mean 35.47. SIL minimum is 0.42. These units are not loudness or waveform RMS.

Limitations: Strict ASR WER 100.0%, CER 17.2% (5 word edits). The reference retains its intentional internal-word gap. These machine-text discrepancies do not establish which sounds were pronounced incorrectly.

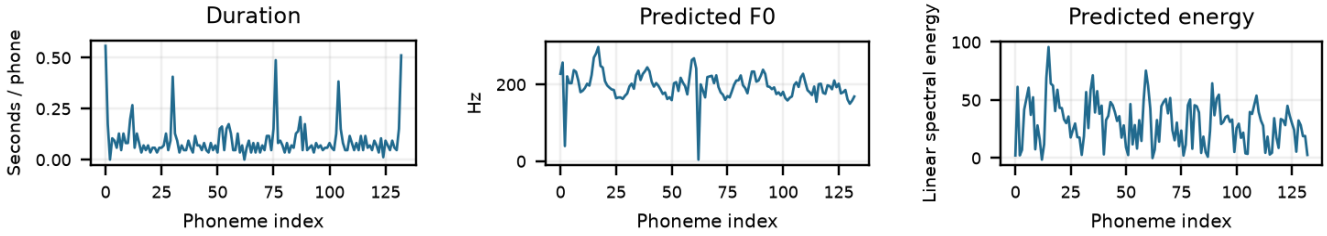
Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/Telugu/test_generated_audios/T3.wav; predictions/T3.npz; variance_adaptor_plots/{duration,pitch,energy}/T3.png.

T4 | Telugu | objective only

Exact Telugu source text: Telugu/test_samples.json, entry T4. The unchanged UTF-8 source is authoritative; complex-script shaping was not validated in this Roman-script PDF.

WAV 12.365 s; RMS -23.94 dBFS; peak 0.373; PCM rail fraction 0.000000. Non-SIL zero-duration phones: 0/119; longest non-SIL duration 0.209 s at phone index 87; longest SIL run 0.557 s.



Non-SIL predicted F0 mean 200.1 Hz; range 149.5-297.4 Hz. Non-SIL predicted energy mean 33.248. Waveform F0 status: measured.

Duration: Total 12.365 s; SIL 3.053 s (24.7%). Longest SIL run 0.557 s at phone index 0. All 2 zero-duration positions are SIL; no speech phone rounds to zero.

Pitch: Non-SIL F0 149.5-297.4 Hz, mean 200.1. The long sentence varies across phones; numeric/date expression and emphasis are not verified by this contour. WORLD voiced-WAV mean 182.9 Hz is a different estimate, not a target error.

Energy: Non-SIL spectral energy 0.99-95.69, mean 33.25. SIL minimum -1.51 is nonphysical after denormalization; the regression is unconstrained. These units are not loudness or waveform RMS.

Limitations: Strict ASR WER 78.6%, CER 47.4%. Reference digits/date/time and hypothesized spoken forms are not canonicalized, so representation differences confound the score. No claim of correct number realization or pronunciation is made.

Listening assessment: not performed. Pronunciation correctness, intelligibility, naturalness, and perceived artifacts are unassessed; tensor and WAV statistics do not establish these properties.

Artifacts: Results/Telugu/test_generated_audios/T4.wav; predictions/T4.npz; variance_adaptor_plots/{duration,pitch,energy}/T4.png.

Plots use the actual phone-level predictions, indexed from zero; each includes SIL entries. Full phoneme labels are available in the corresponding PNG and .npz. Predicted F0 and energy are denormalized with training statistics. The energy panel retains the raw prediction and adds a dashed decoder-conditioning overlay only when the saved SIL control was active. These are not waveform F0 or RMS.

Variance behavior, comparison and limitations

English WAVs average 5.211 s with 6.77% of pooled predicted duration assigned to SIL; Telugu averages 6.334 s with 25.00% SIL. Different text, voices and alignment prevent a controlled language-quality ranking. Final English assignment ASR WER/CER is 27.59%/11.38%, compared with 24.14%/8.98% for original step 7,000/word-level input; stronger fixed-validation performance did not imply uniformly better assignment recognition. Telugu remains 86.49%/24.89%, with substantial word-boundary and written/spoken-form effects. All current non-SIL phones have positive durations and written WAVs have zero PCM rail hits; these are integrity facts, not proof of good speech.

Across five requested samples	English	Telugu
Mean generated duration (s)	5.2106	6.3344
Non-SIL phones rounded to zero (count)	0.0000	0.0000
Longest predicted non-SIL phone (s)	0.2554	0.3483
Maximum PCM rail fraction	0.0000	0.0000

Interpretation limits

The sample sets contain different text and only five sentences per language. Cross-language duration and pitch differences are descriptive, not a controlled language comparison. There is no reference audio for these sentences and no reference-based pronunciation, F0 or duration error. Zero rounded phone durations can omit material, but audibility needs listening. Predicted SIL runs are model decisions, not independently detected waveform silence. PCM rail hits indicate full-scale samples; they cannot recover or prove earlier clipping.

Dataset and supervision limits

English: The 115 source demographic groups are not verified speakers; 114 occur in training and 106 in validation. The final 8,842/context/ordinary-energy recipe is selected for stronger fixed-validation recognition and the input-vowel correction, not for being the lowest assignment-ASR recipe or a quality-gate winner. SIL zeroing was rejected after the controlled assignment regression. Validation includes 8,051 SIL positions among 38,450 phones, including 3,319 zero-duration SIL targets; pitch-loss conventions remain a confound.

Telugu: The female single-voice subset avoids the English identity-proxy issue, but zero-shot Telugu alignment and collapsed acoustic classes remain. Step 2000 is retained after later WER worsens while CER improves. Validation has 4471 SIL positions among 36155 phones, including 1467 zero-duration SIL targets. High source-control WER and word-boundary effects limit ASR interpretation; no native listening was performed.

Svarah conditioning groups can merge voices and are not a verified speaker inventory. Telugu alignment is zero-shot and its acoustic phone mappings collapse distinctions. Confidence and maximum-phone-length filters can reject correct speech or retain incorrect boundaries. WORLD F0 estimates can be wrong; interpolated training pitch is not a voicing target. Native-listener review, improved speaker labels and a Telugu-trained aligner remain separate evidence needs.

Observed failures

The original frozen checkpoint comparison remains inconclusive and the country-list duration gate remains failed. English step 8,842 ordinary-energy fixed-probe WER/CER is 15.38%/11.19%, better than step 7,000's 38.46%/30.60%. SIL zeroing improved those three probes further to 7.69%/2.99%, but worsened the five assignment prompts to 37.93%/14.97% versus 27.59%/11.38% with it OFF, so the override was rejected. A global duration multiplier was also rejected after both probe error rates worsened. Context corrects the input verb-vowel target, not encoded stress or all audible errors. Raw negative SIL-energy predictions persist in some samples. No threshold, raw prediction or checkpoint validation loss was rewritten to conceal these failures.

Next improvements

Obtain native-listener assessment for every exact sentence, including rare words, noun/verb stress, dates and code-switching. Resolve English speaker identities and use a Telugu-trained aligner with checked boundaries. Separate voiced/non-SIL pitch evaluation from SIL/zero-duration conventions, and study unconstrained energy predictions without selecting per-sentence winners. Future generalization claims need new held-out material: these assignment and validation prompts have already informed post-hoc choices.

Reproduction and software environment

Run from the project root with Python 3.11. Use an isolated environment and the pinned requirements. Select the official PyTorch build for the execution host. Obtain gated data using your own authorized account; do not put credentials or raw corpus audio into the public repository.

```
python3.11 -m venv .venv

. .venv/bin/activate

python -m pip install -r requirements.txt

hf auth login

python scripts/download_data.py --help

python scripts/import_parquet.py --help

python scripts/select_candidates.py --language en --manifest data/en/metadata.jsonl --output
data/en/candidates.jsonl

python scripts/select_candidates.py --language te --manifest data/te/metadata.jsonl --output
data/te/candidates.jsonl

python English/preprocess.py --config English/config.yaml --manifest data/en/candidates.jsonl --device mps
--accepted-hours 6.5

python Telugu/preprocess.py --config Telugu/config.yaml --manifest data/te/candidates.jsonl --device mps
--accepted-hours 6.5

python English/train.py --config English/config.yaml

python Telugu/train.py --config Telugu/config.yaml

python English/train.py --config English/config.yaml --resume English/checkpoints/latest.pt

python English/inference.py --config English/config.yaml --checkpoint English/checkpoints/selected.pt --samples
English/test_samples.json --speaker-id <recorded-id>

python Telugu/inference.py --config Telugu/config.yaml --checkpoint Telugu/checkpoints/selected.pt --samples
Telugu/test_samples.json --speaker-id 0

python scripts/analyze_results.py

python scripts/build_report.py --check-only

python scripts/build_report.py

python scripts/validate_submission.py
```

Recorded inference conditioning IDs: English [60]; Telugu [0]. Replace <recorded-id> with the recorded English ID. These are prepared dense IDs. The configuration tables describe checkpoint settings; if a resumed run used a different host or a changed target step count, keep its saved evidence and logs. Use --device cuda or cpu in preprocessing where appropriate.

Recorded environment	English	Telugu
device	cuda:0	cuda:0
torch_version	2.4.1+cu124	2.4.1+cu124
precision	bf16	bf16
parameters	36135745	36103489
batch_size	16	16
median_step_seconds	0.04699426889419556	0.13831993949133903

requirements-lock-macos.txt and preparation_summary.json/extraction_versions record the macOS feature-extraction environment. They do not describe the separate CUDA training host. The table above uses the actual final training invocation's run_summary.json; saved checkpoint configuration supplies the optimizer settings. Package versions absent from those records are not inferred from the local lock. Retain a verified cloud package inventory with the run artifacts for exact environment reconstruction. Cross-device bitwise equality is not promised. Report rendering uses reportlab 5.1.0; waveform analysis runs in the analysis environment.

Sources, licenses and artifact traceability

[1] ming024/FastSpeech2, MIT reference implementation, pinned commit d4e79eb52e8b01d24703b2dfc0385544092958f3

https://github.com/ming024/FastSpeech2/tree/d4e79eb52e8b01d24703b2dfc0385544092958f3

[2] Ren et al. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. ICLR 2021.

https://arxiv.org/abs/2006.04558

[3] Kong et al. HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis. NeurIPS 2020. MIT bundled code.

https://arxiv.org/abs/2010.05646

[4] Javed et al. Svarah: Evaluating English ASR Systems on Indian Accents. INTERSPEECH 2023, pp. 5087-5091. Dataset: CC BY 4.0.

https://huggingface.co/datasets/ai4bharat/Svarah

[5] IIT Madras IndicTTS / SPRINGLab Telugu export. Custom IITM EULA, not generically MIT or CC BY.

https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu

[6] Bournemouth Forced Aligner 2.0.1. GPLv3 package.

https://github.com/tabahi/bournemouth-forced-aligner

[7] CUPE acoustic alignment models. Preserve each model's declared license, including AGPL-3.0 for CUPE-3-mls7.

https://huggingface.co/Tabahi/CUPE-3-mls7

[8] eSpeak NG and Phonemizer pronunciation dependencies. GPL-family licensing.

https://github.com/espeak-ng/espeak-ng

[9] Official IITM IndicTTS EULA URL; official server did not yield the document during the initial audit.

https://www.iitm.ac.in/donlab/indictts/downloads/license.pdf

[10] Reviewed mirror of the primary-authored IITM EULA. Byte identity with the current official file is unverified.

https://huggingface.co/k-m-irfan/Fastspeech2_HS_Flask_API/blob/main/license.pdf

Required corpus notice

COPYRIGHT 2016 TTS Consortium, TDIL, Meity represented by Hema A Murthy & S Umesh, DEPARTMENT OF Computer Science and Engineering and Electrical Engineering, IIT Madras. ALL RIGHTS RESERVED. Acknowledge the Speech Technology Consortium, IIT Madras, Bhashini/MeitY, Prof. Hema A Murthy and Prof. S Umesh. Retain the corpus EULA and downstream restrictions. See THIRD_PARTY_NOTICES.md for the complete project notice record and dependency license distinctions.

Hash-bound evidence

English inference selected.pt SHA256: 48a2bc893c5192a7559408d4843a1d2f99b5bfab6196691edb636c6803ea7bac

English minimum-total best.pt SHA256: 17ad591e4ef9e269df9e3da05737fb4ea52d35d1b5eec7f1229ff3231446afcc

English exact test_samples.json SHA256: b1764a90ac2edd1d97bf37fd83c753a63dd2ac325dc31a21753b26ca9a987079

English dataset revision: ebbf7777fe771490696a3f7b007097606fa8c924

Telugu inference selected.pt SHA256: 7f32328011ab47edda7eadc67dc4331ce4e1a9f11a74fdc771b79e638f72f0bf

Telugu minimum-total best.pt SHA256: 7f32328011ab47edda7eadc67dc4331ce4e1a9f11a74fdc771b79e638f72f0bf

Telugu exact test_samples.json SHA256: 9c42693394a9eea8fe953c8c71a3801945235e9baf697c5d5b8acdb0df382339

Telugu dataset revision: 7ab3c18491bef3f5d4e820c2701b5f78f54031d6

Each Results/<language>/analysis.json binds the checkpoint, configuration, exact test text, inference manifest, individual WAVs and raw predictions to hashes. The builder remeasures the stored WAV/prediction statistics and checks current processed-target fingerprints against the production split audit. Rejected cases and threshold settings are preserved in each preparation folder. No private source paths, gated recordings, credentials or assignment PDF are embedded here.